Lessons From Newton’s Laws: Potential and Work

The bottom half of the table, used fifteen times

These fifteen run on the last two rows of the table in **Note 8**: $W_E = qEd = qV$, and $V = W_E/q = Ed$. The ladder in that note is the picture to hold in your head while you do them.

1. Base your answer to the following question on the diagram below which represents two charged spheres, *X* and *Y*.

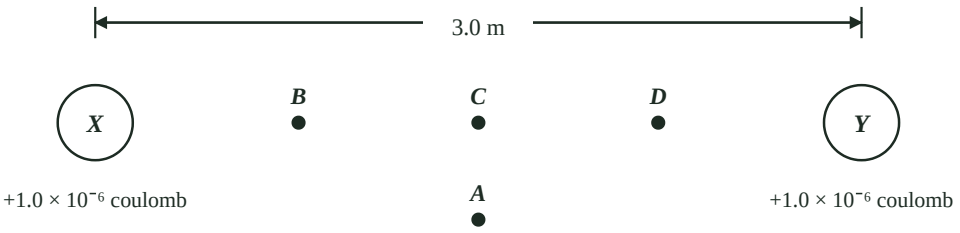


Figure 16. Two like charges 3.0 m apart, with four lettered points.

Moving the two spheres toward each other would cause their electrical potential energy to

A) decrease

B) increase

C) remain the same

2. Base your answer to the following question on the diagram below which represents a system consisting of two charged metal spheres with equal radii.

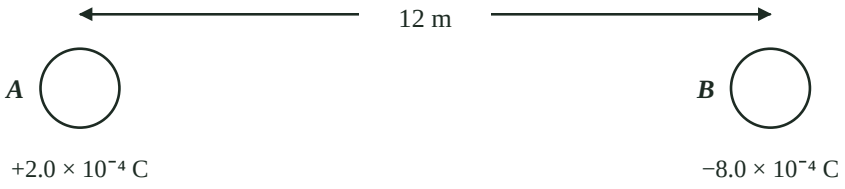


Figure 17. Two unlike charges 12 m apart.

Compared to the electrical potential energy of the system at a separation of 12 meters, the electrical potential energy of the system at a separation of 6 meters is

A) less

B) greater

C) the same

ANSWER THIS ONE YOURSELF

His mark on this item is **boxed and struck through at the same time**, so there is no answer here that can honestly be called his. Work it out instead, and notice what the question turns on: **the two charges have opposite signs.** What does that do to the sign of q_1q_2, and so to the potential energy, as r is halved?

3. Base your answer to the following question on the accompanying diagram which represents two large parallel plates which are oppositely charged. *A*, *B*, and *C* are reference points.

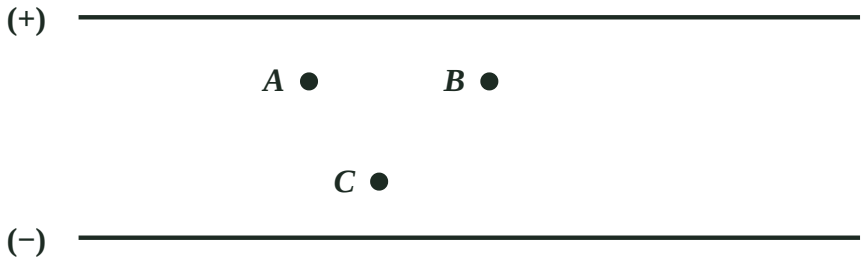


Figure 18. Two oppositely charged plates with three reference points.

If an electron is moved from point *A* to point *C*, the potential energy of the electron will

A) decrease

B) increase

C) remain the same

4. Which electrical unit is equivalent to one joule?

A) volt per meter

B) ampere•volt

C) volt per coulomb

D) coulomb•volt

5. A potential difference of 10.0 volts exists between two points, *A* and *B*, within an electric field. What is the magnitude of charge that requires 2.0×10^{-2} joule of work to move it from *A* to *B*?

6. The graph below shows the relationship between the work done on a charged body in an electric field and the net charge on the body.

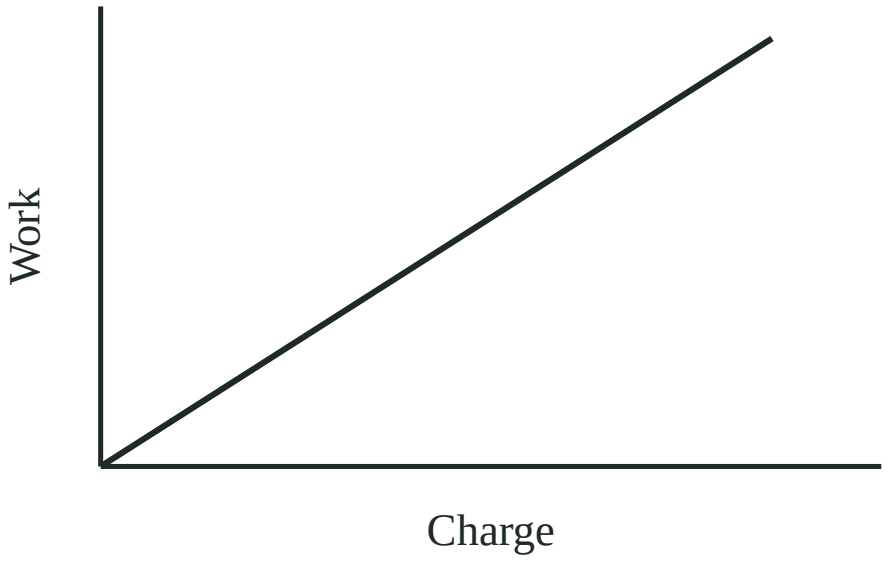


Figure 19. Work against charge.

What does the slope of this graph represent?

A) power

B) potential difference

C) force

D) electric field intensity

7. A volt is defined as a

A) joule/coulomb

B) joule/second

C) coulomb/second

D) joule-second/coulomb

8. The work required to move a charge of 3.0 coulombs through a potential difference of 12 volts is equal to

Two plates and a 500-volt source

Base your answers to questions **9** through **11** on the diagram below which shows two parallel metal plates *A* and *B* connected to a voltage source.

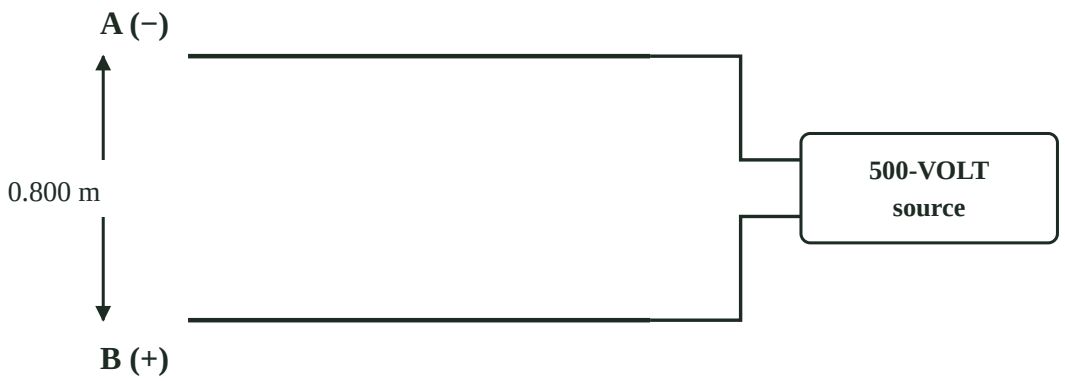


Figure 20. Plate *A* negative, plate *B* positive, 0.800 m apart, across a 500-volt source.

9. If an object carrying a charge of 2 elementary charges starts from rest at plate *A* and is accelerated to plate *B* by the electric field, the total work done by the electric field on the object is

   A) 200. eV

   B) 250. eV

   C) 500. eV

   D) 1,000 eV

10. Convert your answer to joules. Show work.

11. If the mass of the object is 4.0×10^{-26} kg, how fast is it moving?

12. What is the total amount of work required to move a proton through a potential difference of 100. volts?

A) 1.60×10^{-21} J

B) 1.60×10^{-17} J

C) 1.00×10^{2} J

D) 6.25×10^{20} J

13. Convert your answer to electron-volts (eV).

14. Moving 4.0 coulombs of charge through a circuit requires 48 joules of electric energy. What is the potential difference across this circuit?

15. If instead 8.0 coulombs was brought into the field, how would the electric potential change, if at all?

Looking ahead

That is the parallel used up. **Note 11** stops using field and potential and asks what they actually mean — what a field map is really telling you, and why potential, which has no direction at all, is so much easier to work with than the field it came from.